Sahaj Bhadja

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Summary

Software engineer experienced in **Java, C++, Python, React, CSS, Linux, and Windows**, supporting Agile Scrum development, unit and integration testing, and software documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built Python FastAPI services and PostgreSQL schemas for a multi-tenant platform serving 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query optimization end to end.
- Implemented cloud data pipelines processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases, achieving 99%+ reliability across testing and deployment phases for recurring integrations at production scale.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Designed React interfaces and Python Flask backends for an AI grading product serving 150+ instructors, processing 10,000+ daily requests with MySQL persistence and stable API performance at scale.
- Implemented automated unit tests and documented fallback paths for OCR and API failures, reducing customer feedback turnaround time 60% during system integration and validation reliably at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered object-oriented Java and Python microservices with 20+ unit and integration tests using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS in Agile Scrum at scale.
- Optimized Java API integrations using OOP patterns, AWS Glue, and S3, improving response times 30% while producing CSV test reports and failure logs for deployment quality at scale.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a React resource management platform with REST APIs and eight microservices, syncing two million records while reducing resource wastage 42% and increasing conversions 15% at scale.
- Participated in code reviews and Jenkins-based unit and integration testing, cutting deployment time 50% while using Sentry to debug production issues in Agile Scrum delivery at scale.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Translated client requirements into Flask web services and React Native booking modules, mapping 12+ integration points and presenting architecture documentation before implementation for end users at scale.
- Delivered Python Flask and Firebase solutions across 50,000+ records, validating implementation against specifications while achieving sub-200-millisecond responses and reducing manual refund handling 70% for customers at production scale.

Skills

Languages: Java, C++, Python, JavaScript, TypeScript, SQL, Shell
Backend & APIs: Flask, FastAPI, Node.js, Express.js, REST APIs, Spring Boot, Maven
Frontend: React, CSS, HTML, Next.js, TailwindCSS
Databases: PostgreSQL, MySQL, MongoDB, Firebase
Cloud & Systems: Linux, Windows, Docker, Kubernetes, AWS, Azure, Git, Jenkins, Bash
Testing & Practices: Unit Testing, Integration Testing, PyTest, Mockito, JIRA, Agile, Scrum, CI/CD

Projects

CPU-scheduling Simulator Live | *C++, Object-Oriented Programming*
- Implemented object-oriented C++ scheduling algorithms including FCFS, SJF, SRT, and Round Robin with React and CSS controls, reducing average turnaround time 25% across 1,000+ simulated process combinations.

Chat2SQL Code | *Java, Linux, Windows*
- Built Java and TypeScript services supporting PostgreSQL and MySQL, configuring Docker workflows across Windows, Linux, and macOS with schema validation and reliable cross-platform integration end to end at scale.

HMIS - Hospital Management Information System Live | *React, CSS, Web Development*
- Translated healthcare workflow requirements into TypeScript, React, and TailwindCSS appointment modules, adding Sentry monitoring while reducing record retrieval time 50% and administrative workload 30% for operational teams.